

# Kevin Immanuel

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## CAREER OBJECTIVE

Detail-oriented Computer Science student with a passion for building applications from the ground up. Combines full-stack web development expertise with a deep interest in systems architecture, advanced Linux environments, networking, and cryptography. Actively focused on developing smart edge solutions utilizing AIoT.

## EDUCATION

2023 – 2027 B.E., Computer Science & Engineering at **Presidency University**, Bangalore  
2023 Senior Secondary (XII), CBSE Science at **Kendriya Vidyalaya No. 2**, Jalahalli East 82.00%

## WORK EXPERIENCE

**Software Development Intern — Bharat Electronics Limited (BEL)** Aug 2025 – Present  
– Automated PDF-to-software-design document conversion with 90%+ accuracy, featuring dynamic architecture diagrams.  
– Delivered end-to-end full-stack AI solutions for enterprise document processing use cases.

**Web Developer — Co Lead, VIKASANA** Dec 2023 – Present  
– Directed web development and mentored teams for student-led R&D projects, hackathons, and technical workshops.

## PROJECTS

**Automated Billing System (automated-billing-iot)** Python, YOLOv8, FastAPI, Raspberry Pi, IoT  
ML-powered retail solution using Raspberry Pi and YOLOv8 for real-time object detection and weight processing. Frontend built with tkinter GUI; backend runs a custom-trained YOLOv8 model on a FastAPI server. Won Top 5/70 projects at the university-level expo.

**Beat Management System for State Police (police-beat-management)** Next.js, Node.js, WebSockets, MongoDB  
Comprehensive real-time dispatch and incident reporting system for state police, featuring SOS functionality, AI analytics and suggestions, and beat management. Built on the MERN stack with WebSockets and Next.js. Secured Top 10 among 400 teams at Presidency University hackathon.

**6G Protocol Parser (6g-protocol-parser)** Rust, Nom, Networking, Systems  
High-throughput, zero-copy network protocol parser built in Rust using the Nom parsing framework. Features custom bit-level manipulation for 6G-ready packet headers.

**C REST Server (c-rest-server)** C, Socket Programming, Systems, Concurrency  
Low-level REST API server implemented from scratch in C using socket programming and POSIX threads. Handles raw HTTP requests and responses without any external frameworks.

## SKILLS

|                  |                                                                                      |
|------------------|--------------------------------------------------------------------------------------|
| Languages        | Java, Python, JavaScript, TypeScript, C, Go, Rust                                    |
| Web Technologies | React, Next.js, Express.js, Node.js, PHP, Svelte, Flask, Django, WebSockets, FastAPI |
| Databases        | MongoDB, PostgreSQL, SQLite, Redis                                                   |
| Cloud            | Docker, K8s, GCP, Azure, Supabase                                                    |
| Hardware         | Raspberry Pi, ESP32, FPGA                                                            |
| Machine Learning | YOLOv8, Object Detection, Computer Vision, Mistral Basic OCR                         |

## EXTRACURRICULAR ACTIVITIES

**Hackathons** Active participant in various hackathons with consistent top-tier performance, including university-level and national-level competitions.  
**Leadership** Conducted multiple hackathons and workshops through college club both as volunteer and co-leader at VIKASANA.